

Executive Director's Interview Assessment

From the perspective of an Executive Director (ED) on Nomura's Cash Equities Central Risk Book (CRB) desk, testing your mastery of stochastic calculus—specifically the Martingale Representation Theorem (MRT)—is not an academic exercise in pure mathematics. In a modern institutional CRB environment handling algorithmic execution residuals, internal block crosses, and continuous portfolio optimization, the MRT serves as the exact structural justification for the desk's risk-neutral pricing engines and automated delta-hedging infrastructure.

An ED will immediately evaluate your response based on your ability to bridge high-level measure theory with operational execution reality on the desk:

1. **Replication under Execution Friction:** The theoretical MRT assumes a continuous, frictionless market where you can rebalance your delta-hedging positions infinitely. In cash equities, real-world constraints such as market impact, transaction costs, discrete trading intervals, and liquidity gaps make perfect replication impossible. Your response must address how the continuous process ϕ_t transforms into a discrete-time optimization problem under a local risk-minimization framework.
2. **The Change of Measure ($\mathbb{P} \rightarrow \mathbb{Q}$):** The ED will check if you can cleanly differentiate between the physical statistical measure $\mathbb{P}$ (where the strategy experiences historical alpha and drift) and the risk-neutral pricing measure $\mathbb{Q}$ (where the discounted asset paths are structural martingales used to lock in funding costs).
3. **The Multi-Asset Reality:** Cash equity books do not hedge single options against single underlying stocks. You are hedging high-dimensional option portfolios or statistical arbitrage vectors against a basket of correlated underlyings, index futures, and liquid exchange-traded funds (ETFs). The multidimensional extension of the MRT is the foundation for defining your cross-asset hedging matrix.

1. Deep-Dive Mathematical Derivation

Let us derive the martingale property of compensated squared Brownian motion and unpack the Martingale Representation Theorem step-by-painstaking-step.

Step 1: Definition and Conditions of a Martingale Process

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfies the usual conditions (completeness and right-continuity). A continuous stochastic process M_t is defined as a martingale with respect to the filtration $\{\mathcal{F}_t\}$ under the probability measure $\mathbb{P}$ if it rigorously satisfies three distinct axioms:

1. **Adaptation / Measurability:** For every $t \geq 0$, M_t is $\mathcal{F}_t$ -measurable. This implies that at time t , the realization of M_t contains no future information.
2. **Integrability:** For every $t \geq 0$, the expected absolute value of the random variable is bounded:

$$\mathbb{E}^{\mathbb{P}}[|M_t|] < \infty$$

3. **The Martingale Property:** For any two time horizons s and t such that $0 \leq s \leq t$, the conditional expectation of the future value given the current information filtration exactly equals the current value:

$$\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_s] = M_s \quad \text{almost surely}$$

Step 2: Step-by-Step Proof that $W_t^2 - t$ is an $\mathcal{F}_t$ -Martingale

Let W_t be a standard 1-dimensional Brownian motion under $\mathbb{P}$. We wish to prove that the process $M_t = W_t^2 - t$ satisfies the martingale property.

Fix two distinct time points s and t such that $0 \leq s \leq t$. We evaluate the conditional expectation $\mathbb{E}[M_t \mid \mathcal{F}_s]$ component by component.

1. Decomposing the Horizon State using Independent Increments

A fundamental property of Brownian motion is that its increments over non-overlapping intervals are independent of the historical filtration. We break down W_t by anchoring it to its value at time s :

$$W_t = W_s + (W_t - W_s)$$

Substituting this identity into the squared term of our target conditional expectation yields:

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = \mathbb{E}[(W_s + (W_t - W_s))^2 - t \mid \mathcal{F}_s]$$

2. Expanding the Quadratic Algebraic Form

Expanding the polynomial inside the conditional expectation bracket:

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = \mathbb{E}[W_s^2 + 2W_s(W_t - W_s) + (W_t - W_s)^2 - t \mid \mathcal{F}_s]$$

By the linearity property of the conditional expectation operator, we distribute the expectation across each individual scalar term:

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = \mathbb{E}[W_s^2 \mid \mathcal{F}_s] + \mathbb{E}[2W_s(W_t - W_s) \mid \mathcal{F}_s] + \mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s] - \mathbb{E}[t \mid \mathcal{F}_s]$$

3. Simplifying the Expectations Component by Component

We evaluate each of the four distributed terms systematically:

- **Term 1:** $\mathbb{E}[W_s^2 \mid \mathcal{F}_s]$ Since W_s is fully determined at time s , it is $\mathcal{F}_s$ -measurable. It behaves as a known constant under this conditioning restriction:

$$\mathbb{E}[W_s^2 \mid \mathcal{F}_s] = W_s^2$$

- **Term 2:** $\mathbb{E}[2W_s(W_t - W_s) \mid \mathcal{F}_s]$ Using the "taking out what is known" property, the $\mathcal{F}_s$ -measurable multiplier $2W_s$ factors directly outside the conditional expectation operator:

$$\mathbb{E}[2W_s(W_t - W_s) \mid \mathcal{F}_s] = 2W_s \mathbb{E}[W_t - W_s \mid \mathcal{F}_s]$$

Because the forward increment $(W_t - W_s)$ is independent of the past historical filtration $\mathcal{F}_s$, its conditional expectation simplifies to its unconditional expectation. Standard Brownian motion has zero drift, so $\mathbb{E}[W_t - W_s] = 0$:

$$2W_s \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] = 2W_s \cdot 0 = 0$$

- **Term 3:** $\mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s]$ By the same independent increments property, this conditional expectation reduces to the unconditional variance of the forward increment. The variance of a standard

Brownian motion increment scales linearly with the elapsed time interval:

$$\mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s] = \mathbb{E}[(W_t - W_s)^2] = \text{Var}(W_t - W_s) = t - s$$

- **Term 4:** $\mathbb{E}[t \mid \mathcal{F}_s]$ Since time t is a deterministic scalar constant, its expectation is simply itself:

$$\mathbb{E}[t \mid \mathcal{F}_s] = t$$

4. Reassembling the Terms

Substituting these simplified components back into our full distributed equation:

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 + 0 + (t - s) - t$$

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s = M_s \quad \blacksquare$$

This confirms that the compensated squared Brownian motion is a structural martingale.

Step 3: Theoretical Foundation of the Martingale Representation Theorem (MRT)

Let $W_t = (W_t^1, \dots, W_t^d)^\top$ be a d -dimensional standard Brownian motion on our filtered probability space, and let $\{\mathcal{F}_t\}$ be the internal filtration generated explicitly by this process.

Theorem Statement: Let M_t be any sample path continuous, square-integrable martingale with respect to the filtration $\{\mathcal{F}_t\}$ under the probability measure $\mathbb{Q}$, meaning $\mathbb{E}^{\mathbb{Q}}[M_t^2] < \infty$ for all $t \geq 0$. Then, there exists a unique, predictable, and progressively measurable row vector process $\phi_t = (\phi_t^1, \dots, \phi_t^d)$ satisfying the integrability condition $\int_0^t \|\phi_s\|_2^2 ds < \infty$ almost surely, such that for all $t \geq 0$:

$$M_t = M_0 + \int_0^t \phi_s d\mathbf{W}_s^{\mathbb{Q}}$$

Or in differential form:

$$dM_t = \phi_t d\mathbf{W}_t^{\mathbb{Q}}$$

The depth of this theorem cannot be overstated: it states that any complex, square-integrable martingale—no matter how non-linear or path-dependent its final payoff—can be represented entirely as a continuous stochastic integral with respect to the underlying Brownian motion driving the filtration.

Step 4: The Role of MRT in Derivative Pricing and Replication

Let us analyze how the MRT guarantees that options and derivatives can be perfectly replicated in a risk-neutral market.

Consider an equity asset whose price process S_t satisfies the stochastic differential equation (SDE) under the physical measure $\mathbb{P}$:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t^{\mathbb{P}}$$

By Girsanov's Theorem, we can switch to the risk-neutral pricing measure $\mathbb{Q}$. Under $\mathbb{Q}$, the drift of the asset is adjusted to equal the risk-free funding rate r_t . The asset SDE becomes:

$$dS_t = r_t S_t dt + \sigma_t S_t dW_t^{\mathbb{Q}}$$

Let $G_t = e^{-\int_0^t r_u du}$ represent the continuous discount factor process. We apply Ito's Lemma to derive the differential form of the discounted asset price process, $d(G_t S_t)$:

$$d(G_t S_t) = -r_t G_t S_t dt + G_t dS_t = -r_t G_t S_t dt + G_t (r_t S_t dt + \sigma_t S_t dW_t^{\mathbb{Q}}) = \sigma_t G_t S_t dW_t^{\mathbb{Q}}$$

Since this discounted asset process has zero drift under $\mathbb{Q}$, it is a structural $\mathbb{Q}$ -martingale.

Now, consider a path-dependent cash equity derivative or execution performance milestone that delivers a non-linear payoff H_T at maturity T . We define the value process of this derivative at any intermediate time t as its conditional expectation under the risk-neutral measure:

$$V_t = e^{\int_0^t r_u du} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_u du} H_T \mid \mathcal{F}_t \right]$$

Let us analyze the discounted derivative value process, $M_t = G_t V_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_u du} H_T \mid \mathcal{F}_t \right]$. By the Law of Total Iterated Expectations, this conditional expectation process is a square-integrable $\mathbb{Q}$ -martingale.

Because M_t is a $\mathbb{Q}$ -martingale, the **Martingale Representation Theorem** guarantees the existence of an adapted hedging process ϕ_t such that:

$$G_t V_t = V_0 + \int_0^t \phi_s dW_s^{\mathbb{Q}} \implies d(G_t V_t) = \phi_t dW_t^{\mathbb{Q}}$$

To construct a replicating portfolio, we define a self-financing trading strategy consisting of Δ_t shares of the underlying stock S_t and an allocation B_t in the risk-free cash account. The total value of this replicating portfolio is $\Pi_t = \Delta_t S_t + B_t$. The discounted portfolio value process evolves as:

$$d(G_t \Pi_t) = \Delta_t d(G_t S_t) = \Delta_t \sigma_t G_t S_t dW_t^{\mathbb{Q}}$$

To ensure that our replicating portfolio perfectly tracks the derivative value across all sample paths, we equate the two differentials, $d(G_t \Pi_t) = d(G_t V_t)$:

$$\Delta_t \sigma_t G_t S_t dW_t^{\mathbb{Q}} = \phi_t dW_t^{\mathbb{Q}}$$

Matching the stochastic diffusion terms isolates the exact formula for the **Delta Hedging Strategy**:

$$\Delta_t = \frac{\phi_t}{\sigma_t G_t S_t} = \frac{\phi_t e^{\int_0^t r_u du}}{\sigma_t S_t}$$

The MRT provides the theoretical foundation for this result: it guarantees the existence of the process ϕ_t , which allows us to find the exact asset allocation Δ_t needed to perfectly hedge the option's risk across all market states.

2. Application of Theory & Code Portions

Replicating Option Payoffs via Monte Carlo Simulation

While the MRT guarantees that a replicating portfolio exists, calculating the hedging parameters for complex, path-dependent options often requires numerical simulation. In a production CRB system, we use Monte Carlo

simulation to generate asset price paths under the risk-neutral measure $\mathbb{Q}$.

To determine the delta hedging parameter numerically across these simulated paths, we apply the **Malliavin Calculus / Likelihood Ratio Method**. This technique computes the derivative of the option's expected payoff with respect to the initial asset price:

$$\Delta_0 = \frac{\partial V_0}{\partial S_0} = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[H(S_T) \cdot \frac{W_T}{S_0 \sigma T} \right]$$

This formulation allows the trading system to calculate exact delta hedges for digital options, barrier options, or execution algorithms without relying on unstable finite-difference grid lookups.

3. Production-Grade Institutional Implementation

The python engine below implements a high-performance simulation platform. It generates risk-neutral asset paths, prices path-dependent derivatives, and calculates robust delta hedging allocations using the stochastic likelihood ratio framework.

```
# =====
# crb_mrt_hedging_engine.py
#
# Production-Grade Risk-Neutral Option Simulation and Delta Hedging Engine.
# Evaluates path dependent payoffs and computes delta hedges via MRT foundations.
# =====

from __future__ import annotations

import logging
from dataclasses import dataclass
from typing import Final

import numpy as np
import scipy.stats as stats

# -----
# Setup & Constants
# -----
LOG_FMT: Final[str] = "%(asctime)s | %(levelname)-8s | %(name)s | %(message)s"
logging.basicConfig(level=logging.INFO, format=LOG_FMT)
logger = logging.getLogger(__name__)

@dataclass(frozen=True, slots=True)
class PriceHedgingResult:
    """Container holding derivative price metrics and corresponding delta hedge allocations."""
    derivative_price: float
    delta_hedge_allocation: float
    standard_error: float
```

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class CRBMartingaleHedgingEngine:
    """Simulation engine for pricing path-dependent cash equity derivatives."""

    def __init__(self, risk_free_rate: float, volatility: float) -> None:
        self._r = risk_free_rate
        self._sigma = volatility

    def price_and_hedge_asian_call(
        self,
        s0: float,
        strike: float,
        time_to_maturity: float,
        num_paths: int = 100000,
        num_steps: int = 252
    ) -> PriceHedgingResult:
        """Prices an Asian geometric-mean call option and calculates its delta
        hedge allocation.

        Uses the Likelihood Ratio Method to estimate the delta hedge parameter.
        """
        T = time_to_maturity
        dt = T / num_steps

        # Pre-allocate random innovations array to optimize memory layout
        # (num_steps, num_paths) configuration for efficient sequential processing
        Z = np.random.normal(0.0, 1.0, size=(num_steps, num_paths))

        # Initialize paths array at the starting stock price s0
        S = np.zeros((num_steps + 1, num_paths), dtype=np.float64)
        S[0, :] = s0

        # Simulate asset paths under the risk-neutral measure Q using Geometric
        # Brownian Motion
        #  $dS_t = r * S_t * dt + \sigma * S_t * dW_t$ 
        for t in range(1, num_steps + 1):
            drift_step = (self._r - 0.5 * (self._sigma ** 2)) * dt
            diffusion_step = self._sigma * np.sqrt(dt) * Z[t-1, :]
            S[t, :] = S[t-1, :] * np.exp(drift_step + diffusion_step)

        # Calculate the geometric mean across the simulated time paths
        # Geometric Mean =  $\exp( (1 / N) * \sum(\log(S_t)) )$ 
        geometric_mean_paths = np.exp(np.mean(np.log(S[1:, :]), axis=0))

        # Evaluate the Asian call option payoff at maturity:  $\max(\text{Mean} - \text{Strike}, 0)$ 
        payoffs = np.maximum(geometric_mean_paths - strike, 0.0)
        discounted_payoffs = np.exp(-self._r * T) * payoffs

        # Calculate the derivative value as the sample mean of the discounted
        # payoffs
        option_price = float(np.mean(discounted_payoffs))
        stderr_price = float(stats.sem(discounted_payoffs))

        # Calculate the terminal standard Brownian motion value  $W_T$ 
        #  $W_T$  is the cumulative sum of the individual innovations scaled by

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```

sqrt(dt)
    W_T = np.sum(Z, axis=0) * np.sqrt(dt)

    # Apply the Likelihood Ratio Method to calculate the initial delta hedge
    allocation
    # Delta = exp(-r*T) * E[ Payoff * (W_T / (S_0 * sigma * T)) ]
    likelihood_ratio_weight = W_T / (s0 * self._sigma * T)
    delta_estimates = discounted_payoffs * likelihood_ratio_weight
    delta_hedge = float(np.mean(delta_estimates))

    logger.info("Simulation completed. Option Price: %.4f, Calculated Delta
Hedge: %.4f", option_price, delta_hedge)

    return PriceHedgingResult(
        derivative_price=option_price,
        delta_hedge_allocation=delta_hedge,
        standard_error=stderr_price
    )

# -----
# Validation Entrypoint
# -----
if __name__ == "__main__":
    np.random.seed(42)

    # Define market environment parameters
    spot_context = 100.0
    strike_context = 98.0
    tenor_context = 0.5          # 6-month option maturity window
    funding_rate = 0.05          # 5% risk-free financing cost
    asset_volatility = 0.25      # 25% annualized equity volatility parameter

    # Run the pricing and hedging engine
    engine = CRBMartingaleHedgingEngine(risk_free_rate=funding_rate,
volatility=asset_volatility)
    result = engine.price_and_hedge_asian_call(
        s0=spot_context,
        strike=strike_context,
        time_to_maturity=tenor_context,
        num_paths=150000
    )

    print("\n" + "="*65)
    print("  NOMURA CENTRAL RISK BOOK – PRICING & REPLICATION LAB  ")
    print("="*65)
    print(f"Option Valuation Metric Type : Asian Geometric Mean Call")
    print(f"Estimated Derivative Price    : ${result.derivative_price:.4f} +/-
{result.standard_error:.4f}")
    print(f"Optimal MRT Delta Allocation : {result.delta_hedge_allocation *
100:.3f}% of Underlying Book")
    print("-"*65)
    print(f"Replicating Portfolio Strategy Setup:")
    print(f"  Buy {result.delta_hedge_allocation:.5f} shares of underlying equity

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per contract.")
print(f" Finance remaining balance of ${result.derivative_price -
(result.delta_hedge_allocation * spot_context):.4f} via cash account.")
print("="*65 + "\n")

```

3. Key Follow-Ups & Diagnostic Questions

Question 1: If the filtration $\{\mathcal{F}_t\}$ is generated by two independent Brownian motions W_t^1 and W_t^2 , but the market contains only a single liquid underlying asset S_t driven by W_t^1 , why does the Martingale Representation Theorem imply that the market is incomplete?

Answer: The Martingale Representation Theorem states that any square-integrable martingale M_t adapted to the joint filtration $\{\mathcal{F}_t\}$ must be represented as a vector combination of integrals across both independent Brownian motion paths:

$$M_t = M_0 + \int_0^t \phi_s^1 dW_s^1 + \int_0^t \phi_s^2 dW_s^2$$

When we evaluate the discounted derivative value process under the risk-neutral measure $\mathbb{Q}$, its evolution equation contains diffusion terms for both independent risk drivers:

$$d(G_t V_t) = \phi_t^1 dW_t^1 + \phi_t^2 dW_t^2$$

However, because the market contains only a single tradeable underlying asset S_t , the discounted value of our self-financing replicating portfolio ($\Pi_t = \Delta_t S_t + B_t$) can only update its exposure along the first Brownian motion axis:

$$d(G_t \Pi_t) = \Delta_t d(G_t S_t) = \Delta_t \sigma_t G_t S_t dW_t^1 + 0 dW_t^2$$

To achieve perfect replication, we try to solve for the hedging parameter Δ_t by matching the stochastic components of the portfolio and the derivative:

$$\Delta_t \sigma_t G_t S_t = \phi_t^1 \quad \text{and} \quad 0 = \phi_t^2$$

If the derivative's value depends on the second risk factor ($\phi_t^2 \neq 0$, which occurs for cross-asset options or volatility-dependent payoffs), the second equation is impossible to solve.

This mismatch means that you cannot perfectly replicate the derivative's cash flows using the available assets. The presence of an un-hedgeable risk factor (ϕ_t^2) introduces intrinsic tracking error, rendering the market incomplete and preventing perfect delta hedging.

Question 2: How does the presence of discrete rebalancing intervals (e.g., rebalancing once per day instead of continuously) alter the martingale replication error?

Answer: In a continuous-time market, the value of a self-financing replicating portfolio tracks the derivative's value perfectly, maintaining a net zero tracking error process. However, when rebalancing occurs on a discrete schedule $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$, the portfolio's tracking error over an intermediate trading interval $\Delta t = t_{k+1} - t_k$ accumulates as a residual stochastic integral:

$$\epsilon_{t_{k+1}} = (G_{t_{k+1}} V_{t_{k+1}} - G_{t_k} V_{t_k}) - \Delta_{t_k} (G_{t_{k+1}} S_{t_{k+1}} - G_{t_k} S_{t_k}) = \int_{t_k}^{t_{k+1}} (\phi_s - \Delta_{t_k} \sigma_s G_s S_s) dW_s^{\mathbb{Q}}$$

Because the hedging parameter is fixed at Δ_{t_k} throughout the interval while the true optimal hedging process ϕ_s continues to fluctuate with the market, the terms inside the integral no longer cancel out.

By the properties of the Ito isometry, the variance of this discrete tracking error over the interval scales quadratically with the rebalancing step size:

$$\mathbb{E}^{\mathbb{Q}} [\epsilon_{t_{k+1}}^2 \mid \mathcal{F}_{t_k}] = \int_{t_k}^{t_{k+1}} \mathbb{E}^{\mathbb{Q}} [(\phi_s - \Delta_{t_k} \sigma_s G_s S_s)^2 \mid \mathcal{F}_{t_k}] ds \approx \frac{1}{2} (\Gamma_{t_k} \sigma_{t_k}^2 S_{t_k}^2)^2 (\Delta t)^2$$

Where $\Gamma = \frac{\partial^2 V}{\partial S^2}$ represents the option's Gamma profile.

This result highlights why discrete rebalancing introduces variance risk to a Central Risk Book. The trading desk cannot eliminate this risk entirely; instead, it must optimize a regularized utility function that balances the cost of option tracking error against the transaction costs incurred by rebalancing more frequently.